COMPLETING THE SQUARE

1. Solve the following equations by completing the square.
   Give your answer correct to 1 decimal place.

   (a) $x^2 + 2x - 2 = 0$ (b) $x^2 - 4x + 2 = 0$

   (c) $y^2 - 3y - 4 = 0$ (d) $y^2 + 9y + 1 = 0$

   (e) $t^2 - 4t - 4 = 0$ (f) $p^2 - 5p + 1 = 0$

2. Solve the following equations by completing the square.
   Give your answer correct to 2 decimal places.

   (a) $2y^2 + 3y - 4 = 0$ (b) $2t^2 - 2t - 3 = 0$

   (c) $2p^2 - 3p - 1 = 0$ (d) $3x^2 + 2x - 3 = 0$

   (e) $15x^2 - 30x + 10 = 0$ (f) $5y^2 = 9y - 1$

   (g) $2x^2 = 11x + 7$ (h) $2x = 8 - 5x^2$

3. Write $x^2 + 4x + 5$ in the form $(x + a)^2 + b$ where a and b are integers.

4. For all values of $x, x^2 - 10x + 7 = (x - p)^2 - q$, Find the value of the constants p and q.

5. Solve the equation $(x - 7)^2 - 5 = 0$, Write your answer in the form $a \pm \sqrt{b}$ where a and b are integers.

6. By completing the square, solve the equation $x^3 + 10x - 3 = 0$. Give your answer in the form $p \pm \sqrt{q}$ where p and q are integers.

7. (a) Write $x^2 + 14x - 9$ in the form $(x + a)^2 + b$ where a and b are integers.

   (b) Hence, or otherwise, write down the coordinates of the turning point of the graph $y = x^2 + 14x - 9$

8. $6x - 7 - x^2 = p - (x + q)^2$ where p and q are integers. Find the value of p and the value of q.

9. (a) Write $x^2 - 6x - 3$ in the form $(x + a)^2 + b$ where a and b are integers
(b) Hence, or otherwise, solve the equation $x^2 - 6x - 3 = 0$
Write your answer in the form $p \pm \sqrt{q}$ where p and q are integers.

10. Write $2x^2 + 8x + 13$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

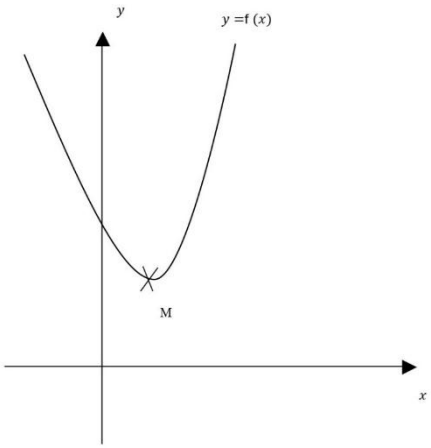
11. (a) Write $2x^2 - 12x + 17$ in the form $a(x - b)^2 + c$ where a, b and c are integers.
(b) Hence, or otherwise, write down the coordinates of the turning point of the graph $y = 2x^2 - 12x + 17$

12. $x^2 + hx + 15 = (x + 3)^2 + k$, Find the value of the constants h and k.

13. $2x^2 - 20x + 10 = p(x - q)^2 - r$, Find the value of the constants p, q and r.

14. The expression $x^2 - 8x + 21$ can be written in the form $(x - a)^2 + b$ for all values of x.

(a) Find the value of a and the value of b.
The equation of a curve is $y = f(x)$ where $f(x)$= $x^2 - 8x + 21$
The diagram shows part of sketch of the graph of $y = f(x)$.
The minimum point of the curve is M.



(b) Write down the coordinates of M.

15. (a) Write $x^2 + 6x + 13$ in the form $(x + a)^2 + b$ where a and b are integers.
(b) Hence, or otherwise, write down the coordinates of the turning point of the graph $y = x^2 + 6x + 13$
(c) Using yours answer to part (b) state the number of solutions to the equation $x^2 + 6x + 13 = 0$
Give a reason for your answer.

16. (a) Write $3x^2 - 12x - 1$ in the form $a(x - b)^2 + c$ where a, b and c are integers.
(b) Hence, or otherwise, find the minimum value of $y = 3x^2 - 12x - 1$
(c) Using your answer to part (b) find the equation of the line of symmetry of the curve $y = 3x^2 - 12x - 1$

17. Write $x^2 + 5x + 9$ in the form $(x + a)^2 + b$.

18. $x^2 + px + 4 = (x + q)^2 - 5$, Find the value of constants p and q.

19. $x^2 + x + p = (x + 2)^2 + 3p$ find the value of p.

20. f(x)= $x^2 - 4x - 5$. The diagram shows a sketch of the graph of y =f (x)

The minimum point of the curve is M.
The points at which the curve crosses the x-axis and the y-axis are P, Q and R.

Find the coordinates of the points M, P, Q and R.

